

1. Title:

EV Charging Station Availability Prediction

2. Dataset:

EV Charging Station Availability Tracking Dataset from Kaggle

3. Dataset Link:

<https://www.kaggle.com/datasets/likithagedipudi/ev-charging-station-availability-tracking>

Rows: 1,317,750 time-series records

Columns: 33 features Size: ~323 MB

Format: CSV with headers

Time Range: July 1 - December 31, 2025

4. Target Variable: EV Charging Station Availability Status (station_status)

5. Task Type: Classification

6. Problem Statement:

The increasing number of electric vehicles creates high demand for efficient charging infrastructure. Predicting charging station availability can help optimize charging operations and improve customer experience.

7. Research Questions:

RQ1. How effectively can baseline machine learning models predict charging station availability?

RQ2. Which machine learning model achieves the best predictive performance?

RQ3. How does preprocessing affect prediction accuracy?

RQ4. Which features most influence charging station availability?

RQ5. How do model rankings vary across evaluation metrics?

RQ6. How robust is the best-performing model under different validation conditions?

RQ7. Which model provides the best trade-off between performance and interpretability?

8. Proposed Methodology:

This study follows a supervised machine learning methodology to predict EV charging station availability using classification models.

Data Collection

The EV Charging Station Availability Tracking dataset will be obtained from Kaggle. The dataset contains operational and temporal information related to EV charging station usage and availability.

Data Preprocessing

The dataset will be cleaned and prepared before model training. The preprocessing steps include:

- Handling missing values
- Removing duplicate records
- Encoding categorical variables
- Feature scaling and normalization
- Outlier treatment
- Train-test data splitting

Exploratory Data Analysis (EDA)

Exploratory analysis will be conducted to understand:

- Feature distributions
- Correlations among variables
- Charging station usage patterns
- Availability class distribution

Visualizations such as histograms, heatmaps, and count plots will be used.

Feature Engineering

Additional features may be generated from the raw dataset, including:

- Peak-hour indicators
- Time-based usage features

- Utilization-related metrics

Machine Learning Model Development

Several supervised learning classification models will be implemented and compared, including:

- Logistic Regression
- Decision Tree
- Random Forest
- Naive Bayes

Model Training and Validation

The dataset will be divided into training and testing sets using stratified train-test splitting. Cross-validation techniques will also be applied to ensure model robustness and generalization.

Model Evaluation

The models will be evaluated using the following classification metrics:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix Analysis

Confusion matrices and ROC curves will also be generated.

Feature Importance and Interpretability

Feature importance analysis will be performed using:

- Random Forest feature importance
- SHAP analysis

This will help identify the most influential factors affecting charging station availability.

Robustness Analysis

The best-performing model will be tested under:

- Different train-test splits

- Cross-validation settings
- Noise injection scenarios
- Missing data simulations

Result Visualization and Reporting

Publication-ready figures and tables will be generated using Python libraries such as Matplotlib and Seaborn. Results will be exported as PDF figures and CSV tables for reporting purposes.

9. Models:

Logistic Regression

Decision Tree

Random Forest

Naive Bayes

10. Best Model:

Based on comparative evaluation, **Logistic Regression** was selected as the final recommended model due to superior predictive performance and practical usefulness.

11. Evaluation Metrics:

Accuracy

Precision

Recall

F1-score

Confusion Matrix Analysis

12. Expected Figures:

RQ1_Baseline_Figure.pdf

RQ2_Model_Comparison_Figure.pdf

RQ3_Preprocessing_Figure.pdf

RQ4_Feature_Importance_Figure.pdf

RQ5_Metric_Sensitivity_Figure.pdf

RQ6_Robustness_Figure.pdf

RQ7_Practical_Usefulness_Figure.pdf

workflow.pdf

dataset_sample.pdf

dataset_structure.pdf

13.Expected Tables:

RQ1_Baseline_Table.csv
RQ2_Model_Comparison_Table.csv
RQ3_Preprocessing_Table.csv
RQ4_Feature_Importance_Table.csv
RQ5_Metric_Sensitivity_Table.csv
RQ6_Robustness_Table.csv
RQ7_Decision_Matrix_Table.csv

14.Deployment and User Interface:

A Gradio-based interactive web application was developed to provide a practical interface for EV charging station recommendation. The system allows users to select date, preferred charging time, and city to receive charging station availability information and the best charging station recommendation based on predicted availability and operational metrics.

Link : <https://huggingface.co/spaces/manideep950/ev-charging-recommendation>